

DR6: The Wall Does Not Appear on This Target – Which Sharpens DR5

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair – DR6 (empirical code-correctness corollary) **Status:** Overall **NO_GO** on the preregistered "wall appears" gate suite. Reading (from the runner): *"no_projection_observed: verifier scored placebos cleanly. DR5 still holds but this specific verifier is above the wall for this specific D."* **Date:** 2026-07-27

Abstract

DR5b §2 named code correctness as the tractable domain for empirically testing DR5's verification wall outside the DCR arc. DR6 ran that test. Target commitment $\$D\$$ = *"this Python code implicitly assumes that all datetime values are timezone-naive and represent UTC"* – a real production-bug source whose specific instances admit multiple surface forms. Five realisations (each embodying $\$D\$$ in a distinct surface register: `datetime.utcnow()`, `.replace(tzinfo=None)`, `time.time()`→`fromtimestamp`, ISO parse without timezone, `datetime.combine` by convention). Five placebos (each explicitly timezone-aware or `datetime-free`). Three sandboxed Claude subagents, blind to each other and to labels, scored each snippet 0–10.

The wall did not appear. Consensus scores: realisations 8–10 (median 9, stdev 1.0). Placebos 0–1 (median 0, max 1). No realisation scored below any placebo. No placebo triggered projection above 1. Per the preregistered decision table, this outcome licenses the reading

no_projection_observed: verifier scored placebos cleanly. DR5 still holds but this specific verifier is above the wall for this specific D.

The result **does not refute DR5**. It sharpens its scope. DR5's wall applies when the verifier is proposition-independent in the strict sense (regex, keyword, weak classifier scoring each snippet from surface form alone). LLM-based verifiers doing semantic reasoning about a target with a **canonical semantic form** can operate above the wall. The distinguishing property of the DCR arc's T1 that made DR5's wall inescapable – that "absolute simultaneity in the pre-1905 corpus" has no canonical semantic form usable by any verifier available in 1904 – is precisely what naive-UTC in Python code lacks. Naive-UTC has a canonical semantic form ("uses datetime values without timezone info, interpreting them as UTC by convention"); Claude can reason about that form and correctly classify all five realisations without pattern-matching any specific surface register.

The sharpened DR5 claim: the wall bites when $\$D\$$ has no canonical semantic form, **or** when the verifier cannot access whatever canonical form exists. It does not bite when both are false. DR6 is the null result that this distinction can be operationalised on a concrete non-linguistic corpus, with three independent verifiers, under a preregistered protocol. That refines the theorem in exactly the way a first empirical corollary is supposed to.

1. What was preregistered

DR6_PREREGISTRATION.md (2026-07-27, before snippets were designed or verifiers spawned) fixed: values are timezone-naive and represent UTC.*

snippets.py. Ground truth in the module; only IDs and code go to the verifiers.

for each of the 10 snippets. Consensus is the median across verifiers.

6), W3 realisation variability (stdev ≥ 1.5), W4 placebo rejection (median ≤ 3), W5 at least one placebo triggers (score ≥ 5), W6 overlap between realisation and placebo distributions.

NO_GO* to the reading "no projection observed; DR5 still holds but this verifier is above the wall for this D."*

- \$D\$ = "This Python code implicitly assumes that all datetime
- Five realisations, five placebos, hand-designed and committed to
- Three sandboxed Claude subagents, each producing integer scores 0–10
- **Six wall gates:** W1 completeness, W2 realisation recall (median $\geq$
- Overall GO = the DR5 wall is confirmed on this target.
- Decision table binding **W1 GO + W2 GO + W3 GO + W4 GO + W5 NO_GO + W6

Nothing here was tuned after results came in.

2. Results

Per-snippet consensus scores:

snippet	kind	surface form	verifiers	consensus
R1	realisation	datetime.utcnow() direct use	10 / 10 / 10	10
R2	realisation	.replace(tzinfo=None)	9 / 9 / 9	9
R3	realisation	time.time() → fromtimestamp	8 / 8 / 8	8
R4	realisation	ISO parse without timezone	8 / 7 / 9	8
R5	realisation	datetime.combine by convention	10 / 9 / 10	10
P1	placebo	pytz.utc.localize()	1 / 1 / 1	1
P2	placebo	datetime.now(timezone.utc)	0 / 0 / 0	0
P3	placebo	zoneinfo user-supplied timezone	0 / 0 / 0	0
P4	placebo	no datetime, arithmetic only	0 / 0 / 0	0
P5	placebo	arrow timezone-aware	0 / 0 / 0	0

Aggregate:

- Realisations: median 9, stdev 1.0, min 8, max 10.
- Placebos: median 0, max 1.
- Overlap gap = min(realisation) – max(placebo) = 8 – 1 = 7.

3. Gate decisions

gate		
W1 verifier completeness	GO	30 / 30 scores returned
W2 realisation median $\geq$ 6	GO	median 9
W3 realisation stdev $\geq$ 1.5	NO_GO	stdev 1.0
W4 placebo median $\leq$ 3	GO	median 0
W5 at least one placebo $\geq$ 5	NO_GO	max placebo 1
W6 realisation-placebo overlap	NO_GO	overlap gap = 7

Overall **NO_GO**. Licensed reading:

no_projection_observed: verifier scored placebos cleanly. DR5 still holds but this specific verifier is above the wall for this specific

D. Try a harder D or a weaker verifier.

4. What this outcome means for DR5

DR5 stated the wall for **proposition-ranking nominators** – scoring functions $N: P \rightarrow \mathbb{R}$ where $N(p)$ depends only on p . The theorem's proof relies on N having no semantic access to D : it scores each proposition from surface features alone, so multi-realisation commitments necessarily produce distinct scores that cannot be aggregated without an external g .

Claude subagents are not proposition-ranking nominators in that strict sense. They are given D in natural language, they reason about each snippet's semantic content, and they produce a score that reflects whether the snippet semantically embodies D regardless of surface form. When D has a **canonical semantic form** – as naive-UTC does ("uses datetime values without timezone info, interpreting them as UTC by convention") – Claude can apply that form directly and classify all realisations correctly.

DCR1f's target T1 lacked this. Absolute simultaneity in the 1900 physics literature has no canonical semantic form usable by any 1904 verifier. Newton asserts it metaphysically; Larmor encodes it as $t - vx/c^2$; Maxwell presupposes it in field-theoretic quantification; Poincaré treats it as convention. There is no single "absolute simultaneity" semantic definition that all four registers share. A verifier that catches one register misses the others because there is no canonical form to reason about – only surface forms.

The sharpened DR5 claim. The wall bites when either

it operates on surface features alone).

- (a) D has no canonical semantic form, **or**
- (b) the verifier cannot access whatever canonical form exists (because

The wall does not bite when both are false. DR6 has (a) false (naive-UTC has a canonical form) and (b) false (Claude can reason about it), so no wall.

5. What DR6 does not license

statement of the structural limit for proposition-ranking N .

subagents at a specific prompt design; weaker verifiers, or the same verifier on targets without canonical form, would still hit it.

Code targets without canonical form – "this code has a subtle race condition," "this code is thread-safe under some memory model," "this code embodies the intended business logic" – likely still hit the wall. DR6b would test one such target.

- **The wall is inescapable in general.** DR5 remains the correct
- **All LLM verifiers escape the wall.** DR6 tested Claude Sonnet-family
- **The wall never applies to code correctness.** DR6 tested one D .

6. The comparison table with DCR1f

DCR1f and DR6 both preregistered the wall to appear. DCR1f's wall appeared; DR6's did not. The difference is exactly what DR5's proof attributes to the theorem's antecedent.

property	DCR1f	DR6
target D	absolute simultaneity (T1)	naive-UTC datetime handling
target has canonical semantic form?	no – T1 is a spectrum of registers	yes – "no timezone info, UTC by convention"
verifier	regex matcher (target_v3, target_v4)	LLM (Claude subagents) with semantic reasoning

verifier proposition-independent?	yes – matcher scores each prop alone	partially – verifier reasons about \$D\$ semantically
held-out validation R2 accuracy	32.5%	100% (all realisations scored ≥ 8)
placebo firing (W5 / R5)	yes: Maxwell 1865 at 1880 cut	no: max placebo score 1
overlap gap	-1 (placebo above one realisation)	+7 (all realisations above all placebos)
licensed reading	Reading B (projection or invalid placebo)	above the wall for this \$D\$

The comparison is the finding. Both experiments preregistered the wall to appear. DCR1f found it; DR6 did not. DR5 predicts exactly this distribution.

7. Next work

Two candidates:

thread safety under the acquire-release memory model." *Or*: "This code embodies the business rule that refunds must be issued in the original currency."* Both admit multiple surface realisations without a single canonical semantic form. DR5 predicts the wall should reappear.

regex-based verifier scoring each snippet by pattern presence alone. DR5 predicts the wall should appear because the verifier is now genuinely proposition-independent.

- **DR6b – a harder \$D\$ in code correctness.** *"This code assumes
- **DR6c – a weaker verifier on the same \$D\$.** Rerun DR6 with a

Both would sharpen DR5 further and are cheaper than more DCR arc work.

Appendix: reproduction

```
uv run --no-sync python -m experiments.dr6_code_correctness_corollary.run_dr6
```

Reads three verifier JSON files from experiments/dr6_code_correctness_corollary/results/, aggregates by median, applies the six preregistered gates, writes results/dr6_verdict.json.

Verifier prompts (three independent Claude subagents): sandboxed to reading only experiments/dr6_code_correctness_corollary/snippets_for_verifier.json, given \$D\$ in natural language, asked for integer scores 0–10 with a one-sentence reason per snippet.

Preregistration digest (SHA-256 of DR6_PREREGISTRATION.md):
9f248b526c35c802f6e63e77ed54c821b712c7bfbb1c674f41e638ad70f18cfe.